



Beatrice Marcu

Gameplay / AI Programmer

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Profile

Gameplay & AI Programmer studying Creative Media and Game Technologies at Breda University of Applied Sciences — a project-driven program where students work in professional, studio-like environments from day one. With a strong interest in building gameplay systems that feel both technically solid and enjoyable to interact with, I focus on figuring out how systems connect together — whether that's AI behaviour, gameplay mechanics, tools, or UI — and turning complex ideas into something clean, responsive, and easy to work with.

Most of my experience comes from working in C++ and Unreal Engine 5, where I have developed systems such as inventory and quest systems, combat AI and team communication AI. I particularly enjoy working on systems-heavy gameplay and AI because it combines problem solving, technical depth, and player experience in a really satisfying way. Alongside programming, I also enjoy collaborating closely with designers and artists to build features that are flexible and practical for the whole team.

Through both personal and team projects, I've gained experience contributing across gameplay, camera systems, controller support, puzzles, and UI while learning how to communicate and iterate effectively within a production environment.

Projects

Inventory System - Custom C++ Engine 2026

Grid-based inventory system with drag-and-drop, stacking, and full save/load

- ECS-EntT Library
- Fully configurable layout — slot groups, grid dimensions, spacing, and icon paths set through ImGui at runtime
- Full JSON serialization for save/load
- Swap inventory visuals entirely by changing texture paths, no code required
- Custom item creation tool for designers (no code required)
- Drag-and-drop with left-click (move all) and right-click (split stack) support
- Multi-digit quantity displays using individual sprite entities
- Item popup system (Use/Equip/Drop functionality)

The Final Print - Unreal Engine 2025

Winner of Best Student Game 2025 at the Dutch Game Awards!

- Multi-disciplinary collaboration
- Unreal Blueprints
- Multiple puzzle types (lock-picking, code-breaking)
- NPC interrogation and dialogue trees
- Photography evidence system

Personal details

Website

beaq15.github.io

LinkedIn

linkedin.com/in/beatrice-marcu-904656262

Software

Microsoft Visual Studio

Unreal Engine

Unity

Github

Perforce

Jira

Skills

C++

C#

HTML

Python

Pseudocode

SDL

OpenGL

Languages

Romanian

English

AI Team Communication Plugin - Unreal Engine

2025

Fact-based AI communication system with realistic constraints

- Fact-based knowledge sharing with confidence decay
- Multi-channel communication routing
- Distance-based communication latency
- Role-based rebroadcasting with hop limits
- Area claim system for coordinated search
- Line-of-sight and audio detection integration
- Weak pointer safety for destroyed receivers

Quest system

2024

2D feature created from scratch while developing a custom engine

- Hierarchical quest structure (Quests → Stages → Objectives)
- Multiple objective types (Kill, Build, Collect, TrainUnits, Location)
- Full serialization support with JSON export/import
- Designer-friendly editor interface
- Polymorphic objective serialization
- Automatic quest progression and UI updates

Marisol

2024

3D team game developed in UE5 as second year students

- Dynamic camera system with smooth transitions
- Combat mechanics with shield defense system
- Player respawn with fade transitions
- Dialogue system with custom font integration
- Third-person action-adventure gameplay

Education

Creative Media and Game Technologies, Programming

Sep 2022 - Present

Breda University of Applied Sciences, Breda, Netherlands

- Game programming
- C++
- Unreal Engine 5

Baccalaureate, Computer Science

Sep 2018 - Jul 2022

Tudor Vianu National College of Computer Science

Subjects taken in the Romanian Baccalaureate:

- Romanian Language and Literature
- Mathematics
- Computer Science

Field(s) of study

- mathematics-informatics, Intensive informatics

Internships

Project Lead Developer

Sep 2023 - Jan 2024

JoTo-Games, Breda, North Brabant, Netherlands

- Sole programmer on the project, responsible for gameplay and technical implementation
- Developed gameplay systems and mechanics using Unity and C#
- Collaborated closely with artists and designers during development
- Contributed to the project from prototyping to final polish

Certificates

Cambridge Certificate of Advanced English

Diploma of good results at the ECDL exams

Diploma of Education

- Computer Essentials Module
- Online Essentials Module
- Word Processing Module
- Spreadsheets Module
- Presentation Module
- Using Databases Module
- Computer Essentials Module
- Online Essentials Module
- Word Processing Module
- Spreadsheets Module
- Presentation Module
- Using Databases Module
- **Skills:** Python (Programming Language)